Introduction to Cryptography

Lecture 15: DES Algorithm

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What is DES?

DES – Data Encryption Standard

- A symmetric-key block cipher.
- Developed from IBM's Lucifer cipher.
- Adopted as a U.S. federal standard in 1977.
- Uses a Feistel network structure.
- Encrypts data in 64-bit blocks.
- Accepts a 64-bit key input.
- Only 56 key bits are effective; 8 bits are used for parity.
- Performs 16 encryption rounds.



DES Parameters

Parameter	DES
Block size	64 bits
Key input	64 bits
Effective key size	56 bits
Number of rounds	16
Round-key size	48 bits
Structure	Feistel network
Number of S-boxes	8
S-box input	6 bits
S-box output	4 bits

DES Encryption: High-Level View



Each round uses a different 48-bit subkey:

$$K_1, K_2, \ldots, K_{16}$$

Initial Permutation (IP)

The first transformation applied to a 64-bit plaintext block is:

Initial Permutation (IP)

- Rearranges all 64 input bits according to a fixed table.
- The permutation is independent of the secret key.
- It produces two 32-bit halves.

$$IP(P) = L_0 \parallel R_0$$

where:

$$|L_0| = 32 \qquad |R_0| = 32$$

$$\text{64-bit input} \xrightarrow{IP} \text{32-bit } L_0 + \text{32-bit } R_0$$

DES Feistel Structure

For round i:

$$\boxed{L_i = R_{i-1}}$$

and

$$\boxed{R_i = L_{i-1} \oplus F(R_{i-1}, K_i)}$$

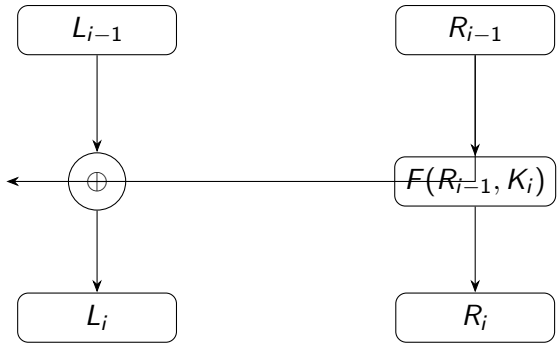
where:

$$i = 1, 2, \ldots, 16$$

and:

$$K_i = 48\text{-bit round key}$$

DES Feistel Structure



One DES Round

Each round performs two operations:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Therefore:

1. The previous right half becomes the new left half.
2. The round function is applied to the previous right half.
3. The round-function output is XORed with the previous left half.

$$\boxed{\text{32-bit } R \rightarrow F \rightarrow \text{32-bit output}}$$

DES Round Function

The round function is:

$$\boxed{F(R, K) = P\left(S\left(E(R) \oplus K\right)\right)}$$

It contains four major stages:

1. Expansion permutation
2. XOR with round key
3. S-box substitution
4. P-box permutation

$$32 \xrightarrow{E} 48 \xrightarrow{\oplus K} 48 \xrightarrow{S} 32 \xrightarrow{P} 32$$

Step 1: Expansion Permutation

The right half entering the round function is:

$$R_{i-1} = 32 \text{ bits}$$

The expansion function E converts it into:

$$E(R_{i-1}) = 48 \text{ bits}$$

Why expansion?

- The DES round key is 48 bits.
- Therefore the data must also be 48 bits before XOR.
- Expansion repeats selected input bits.
- The repeated bits also connect neighboring groups.

$$\boxed{32 \text{ bits} \xrightarrow{E} 48 \text{ bits}}$$

Step 2: XOR with Round Key

The expanded value is:

$$E(R_{i-1}) = 48 \text{ bits}$$

The round key is:

$$K_i = 48 \text{ bits}$$

They are combined using XOR:

$$\boxed{B = E(R_{i-1}) \oplus K_i}$$

Therefore:

$$48 \text{ bits} \oplus 48 \text{ bits} = 48 \text{ bits}$$

$$\boxed{\text{Expanded Data} \oplus \text{Secret Round Key}}$$

Step 3: S-Box Substitution

DES uses:

$$\boxed{8 \text{ S-boxes}}$$

Each S-box performs:

$$6 \text{ bits} \rightarrow 4 \text{ bits}$$

The 48-bit input is divided into:

$$8 \times 6 = 48 \text{ bits}$$

Each S-box produces:

$$8 \times 4 = 32 \text{ bits}$$

Thus:

$$\boxed{48 \rightarrow 32}$$

S-boxes provide the major nonlinear transformation in the DES round function.

DES S-Box Address Calculation

Consider a 6-bit S-box input:

$$b_1 b_2 b_3 b_4 b_5 b_6$$

The row is determined by the first and last bits:

$$\boxed{\text{Row} = b_1 b_6}$$

The column is determined by the middle four bits:

$$\boxed{\text{Column} = b_2 b_3 b_4 b_5}$$

Numerically:

$$\text{Row} = 2b_1 + b_6$$

$$\text{Column} = 8b_2 + 4b_3 + 2b_4 + b_5$$

The selected table entry gives a 4-bit output.

S-Box Numerical Example

Consider:

$$\boxed{101011}$$

First and last bits:

$$b_1 b_6 = 11_2$$

Therefore:

$$\text{Row} = 3$$

Middle four bits:

$$0101_2$$

Therefore:

$$\text{Column} = 5$$

Continue...

Hence the S-box lookup is performed at:

Row 3, Column 5

Note:
The resulting 4-bit value depends on which DES S-box ($S_1, \ldots, S_8$) is being used.

Step 4: P-Box Permutation

After S-box substitution:

$$48 \text{ bits} \rightarrow 32 \text{ bits}$$

The 32-bit result is passed through the DES permutation:

$$\boxed{P\text{-box}}$$

The P-box:

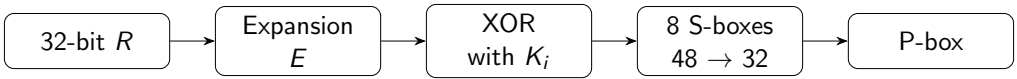
- Rearranges the 32 bits.
- Spreads S-box outputs across later rounds.
- Contributes to diffusion.

Therefore:

$$F(R, K) = P(S(E(R) \oplus K))$$

$$\boxed{32 \rightarrow 48 \rightarrow 48 \rightarrow 32 \rightarrow 32}$$

Complete DES Round Function



$$F(R, K) = P\left(S\left(E(R) \oplus K\right)\right)$$

DES Key Generation

DES starts with a:

$$\text{64-bit key input}$$

Eight bits are parity bits.
Therefore:

$$64 - 8 = 56$$

effective key bits.

The key schedule generates:

$$K_1, K_2, \ldots, K_{16}$$

where:

$$|K_i| = 48$$

$$\boxed{\text{64-bit key} \rightarrow \text{56-bit effective key} \rightarrow 16 \times \text{48-bit round keys}}$$

Permuted Choice 1: PC-1

The first key transformation is:

$$\boxed{\text{PC-1}}$$

PC-1:

- Takes the 64-bit key.
- Removes the eight parity bits.
- Permutes the remaining 56 bits.

The result is divided into two halves:

$$C_0 = 28 \text{ bits}$$

$$D_0 = 28 \text{ bits}$$

Therefore:

$$\boxed{C_0 \parallel D_0 = 56 \text{ bits}}$$

Circular Left Shifts

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Shifts	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

For every round:

$$C_{i-1} \to C_i$$

$$D_{i-1} \to D_i$$

using the specified circular left shift.

Permuted Choice 2: PC-2

After the circular shifts:

$$C_i \parallel D_i$$

contains:

56 bits

PC-2 selects and permutes 48 bits:

$$\boxed{56 \rightarrow 48}$$

Therefore:

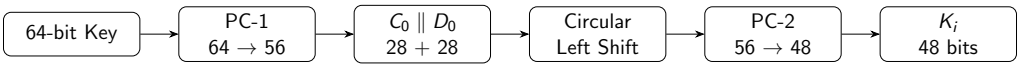
$$\boxed{K_i = PC2(C_i \parallel D_i)}$$

for:

$$i = 1, 2, \ldots, 16$$

Thus:

DES Key Schedule: Complete Flow



The process is repeated for all 16 rounds.

The 16 DES Rounds

After the Initial Permutation:

$$IP(P) = L_0 \parallel R_0$$

Sixteen Feistel rounds are performed:

$$(L_0, R_0) \to (L_1, R_1) \to \cdots \to (L_{16}, R_{16})$$

For each round:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Each round uses a different key:

$$K_1, K_2, \ldots, K_{16}$$

Final Permutation

After Round 16, the two halves are combined in reverse order:

$$R_{16} \parallel L_{16}$$

This is then passed through:

$$\boxed{IP^{-1}}$$

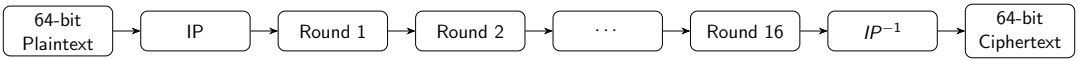
where IP^{-1} is the inverse of the Initial Permutation.
Therefore:

$$\boxed{C = IP^{-1}(R_{16} \parallel L_{16})}$$

The result is the 64-bit ciphertext.

$$\text{64-bit data} \xrightarrow{IP^{-1}} \text{64-bit Ciphertext}$$

Complete DES Encryption Flow



$$IP \rightarrow 16 \text{ Feistel Rounds} \rightarrow IP^{-1}$$

Numerical: DES Key Space

The effective DES key size is:

$$56 \text{ bits}$$

Therefore, the total number of possible keys is:

$$2^{56}$$

$$\boxed{2^{56} = 72,057,594,037,927,936}$$

Approximately:

$$7.2 \times 10^{16}$$

possible keys.

For an ideal exhaustive search, the expected number of trials is approximately:

$$\frac{2^{56}}{2} = 2^{55}$$

Numerical: One DES Round

Suppose:

$$L_{i-1} = \text{32-bit value}$$

$$R_{i-1} = \text{32-bit value}$$

and:

$$K_i = \text{48-bit round key}$$

First calculate:

$$E(R_{i-1}) = 48 \text{ bits}$$

Then:

$$X = E(R_{i-1}) \oplus K_i$$

Next:

$$S(X) = 32 \text{ bits}$$

Security of DES

DES was considered strong when introduced, but its major long-term weakness is the:

56-bit effective key

As computing power increased:

- Exhaustive key search became increasingly practical.
- DES became unsuitable for high-security modern applications.
- Its relatively small key space became a fundamental limitation.

DES has also been extensively studied with respect to:

Differential Cryptanalysis

and:

Linear Cryptanalysis

DES: Historical Significance

DES is historically important because it:

- Became one of the most widely deployed block ciphers.
- Demonstrated the practical use of Feistel networks.
- Provided an important foundation for the study of modern block-cipher design.
- Influenced later standards and cryptographic systems.

The development path can be summarized as:

$$\boxed{\text{DES} \rightarrow \text{Triple DES} \rightarrow \text{AES}}$$

DES vs AES

Feature	**DES**	**AES**
Block size	64 bits	128 bits
Key sizes	56 effective bits	128, 192, 256 bits
Rounds	16	10, 12, 14
Structure	Feistel network	Substitution-Permutation Network
Status	Legacy / historical	Modern standard

AES provides substantially larger key sizes and is suitable for modern cryptographic applications.

Quick Classroom Questions

1. Why does DES use a 64-bit key input but have only 56 effective key bits?
2. Why is the right half expanded from 32 to 48 bits?
3. What is the purpose of the S-boxes?
4. How many round keys are generated in DES?
5. What is the size of each round key?
6. Why does DES use a Feistel structure?
7. Why is DES no longer recommended for new systems?

References

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson.
2. William Stallings, *Computer Security: Principles and Practice*, Pearson.
3. NIST, *Data Encryption Standard (DES)*, FIPS PUB 46-3.
4. NIST, *Data Encryption Standard*, Federal Information Processing Standards Publication.
5. Alfred J. Menezes, Paul C. van Oorschot, and Scott A. Vanstone, *Handbook of Applied Cryptography*, CRC Press.

Primary technical reference: NIST FIPS PUB 46-3

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